# ASTRO BLASTER

## A 3D Shooter Game Using OpenGL

**COURSE CODE & TITLE**

**CSM 3124 - COMPUTER GRAPHICS**

**TEAM NAME**

**ARCADIA**


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# OUTLINES

# INTRODUCTION

- **Language & Tools:** C++, OpenGL
- **Type:** Space Shooter Game
- **Player Role:** Control spaceship, destroy enemies
- **Features:** Smooth animation, collision detection, real-time rendering
- **Purpose:** Apply computer graphics concepts in an interactive way

# MOTIVATION

1. Apply graphics concepts in practice
2. Insprired by ***Galaga*** and ***Space Invaders***
3. Build an expandable expandable framework
4. Gain hands-on experience with OpenGL

# PROBLEM STATEMENT

1. Smooth motion, collision, and explosion effects
2. Multiple enemy types
3. Efficient rendering system
4. Sound, power-ups and slow motion

# OBJECTIVES

- Apply computer graphics concepts in an interactive game
- Learn and implement OpenGL programming techniques
- Develop a lightweight game framework

# PROJECT DESCRIPTION

## USER MODE

Control spaceship using **Left** and **Right** arrow keys

Player has **5 lives**

Collect boosters to gain extra lives, to slow enemies or increase bullets

Survive and destroy enemies to earn higher scores

Every 10 points increases the game level and difficulty

# PROPOSED SYSTEM & SCOPE

## SYSTEM

- C++ and OpenGL-based rendering system
- Real-time event handling through keyboard input
- Uses GLUT for window creation, display, and timer function

## SCOPE

- Introduce AI-based enemies
- Expand for mobile or VR platorms

# PROJECT METHODOLOGY

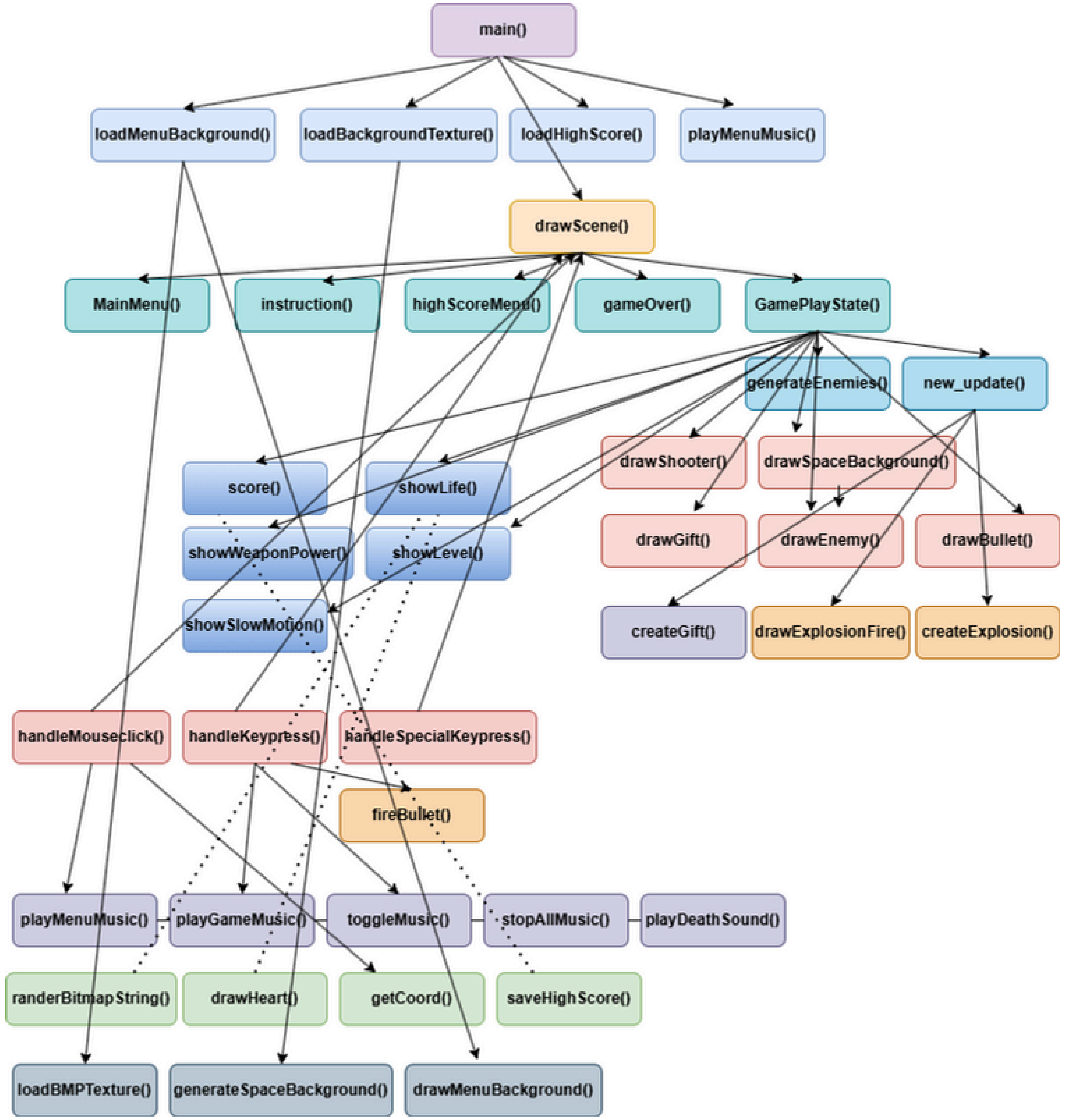


**Fig.1: Project Architecture**

# ACTIVITY DIAGRAM

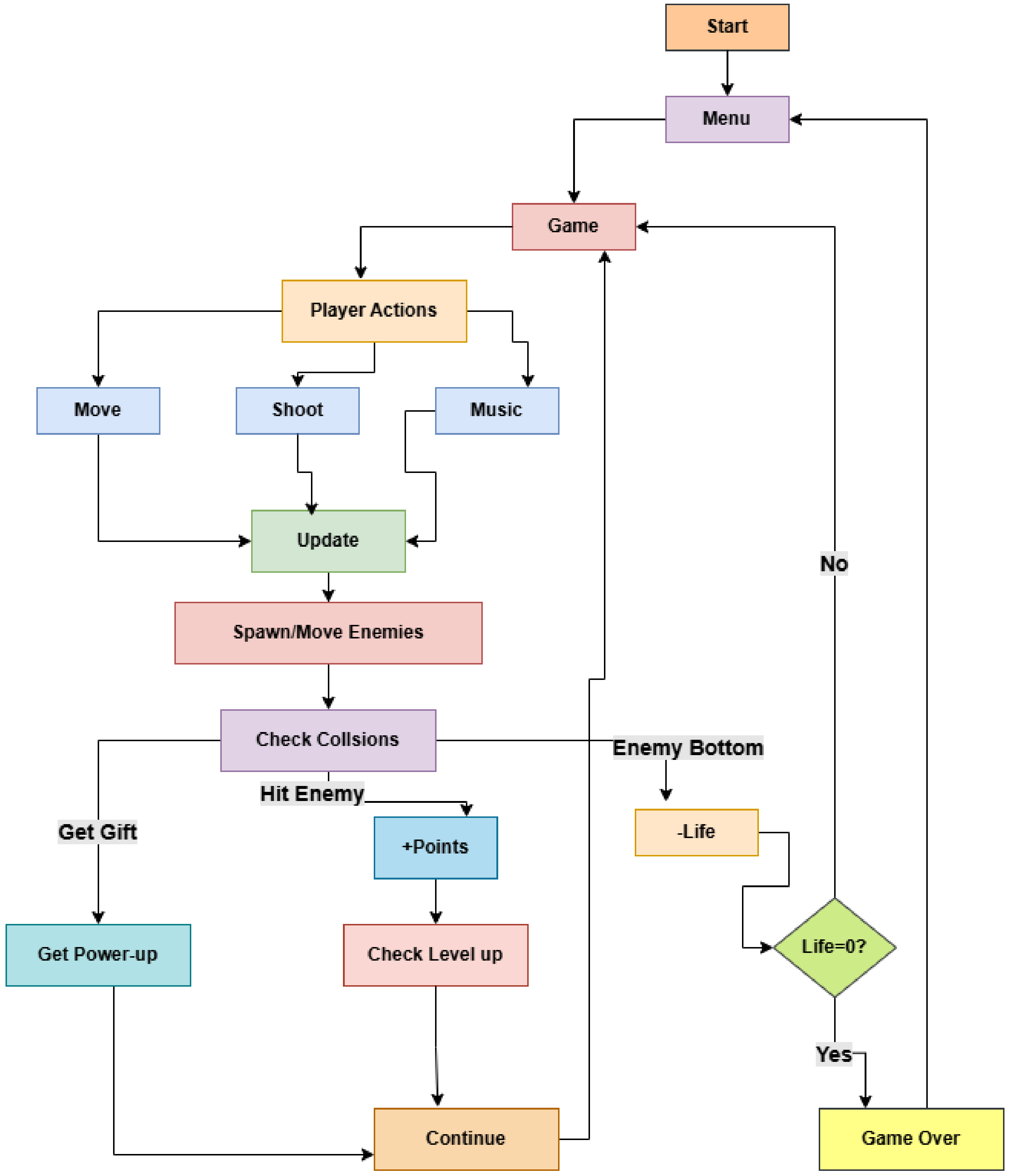


**Fig.2: Activity Diagram**

# IMPLEMENTATION

## Major Functions:

- drawEnemy( )
- drawShooter( )
- drawBullet( )
- createExplosion( )
- generateEnemies( )
- drawScene( )
- handleKeypress( )

- Logic handled in new_update( ) using **GLUT timer functions**
- Sound integrated using **PlaySound API**

# CHALLENGES

1. Lack of engaging 3D examples
2. Faced difficulty while adding background music
3. Multiple sound effects caused conflicts in playback
4. OpenGL (GLUT) lacks advanced shader support

# FINAL OUTPUT


**Fig.3: Main Menu**

# FINAL OUTPUT


**Fig.4: Instructions Interface**

# FINAL OUTPUT

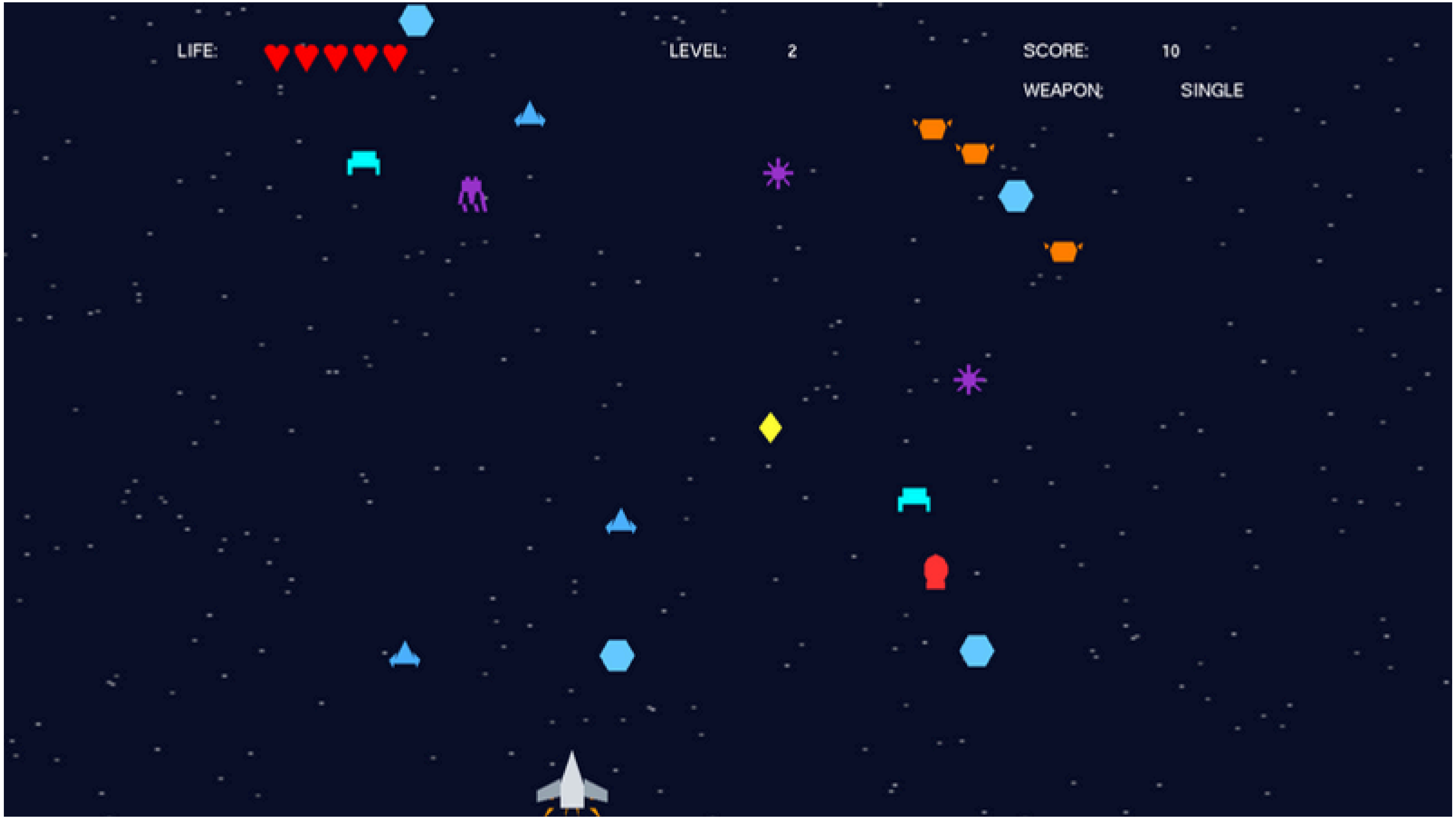


**Fig.5: Game Play Interface**

# FINAL OUTPUT


**Fig.6: Game Over Interface**

# FINAL OUTPUT


**Fig.7: High Score Interface**

# CONCLUSION AND FUTURE WORK

## Conclusion:

- Built an interactive space shooter game
- Gained experience in real-time Graphics Project

## Future Work:

- Add **Boss fights** and **Multiplayer mode**
- Introduce new **weapons** and **AI enemies**
- Optimize for **Mobile/VR plstforms**

# THANK YOU